The Gaussian Moments Conjecture in Two Variables

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23 July 2026

Abstract

The Gaussian Moments Conjecture asks whether a polynomial whose positive powers all have Gaussian integral zero must eventually remain zero after multiplication by any fixed polynomial. We prove the conjecture for two real Gaussian variables. The proof uses circular coordinates to separate rotation from radius. The angular part becomes a constant-term problem, while the radial part becomes the simple functional $U^j \mapsto j!$. An exposed lower face of the resulting Newton diagram, the constant-term theorem of Duistermaat and van der Kallen, and reduction modulo a prime then isolate a nonzero lowest term. This forces every polynomial with vanishing pure moments to have rotational weights of one strict sign, from which the mixed-moment conclusion is immediate. Together with the recent counterexamples of Long in every dimension at least three, this determines the exact dimensional range of the conjecture.

1 Introduction

Let $X_1, \ldots, X_d$ be independent standard real Gaussian random variables. The Gaussian Moments Conjecture, introduced by Derksen, van den Essen, and Zhao [1], is the following statement.

Gaussian Moments Conjecture GMC(d)**.** If $P \in \mathbb{C}[X_1, \ldots, X_d]$ satisfies

$$\mathbb{E}(P^m) = 0 \qquad \text{for every } m \geq 1,$$

then, for every $Q \in \mathbb{C}[X_1, \ldots, X_d]$,

$$\mathbb{E}(QP^m) = 0 \qquad \text{for all sufficiently large } m.$$

The conjecture was motivated in part by its connection with the Jacobian Conjecture [1]. Long recently found explicit counterexamples to GMC(d) for every $d \geq 3$ [3]. The two-variable case was therefore the last undecided dimension.

Theorem 1.1. *The Gaussian Moments Conjecture holds in two real variables.*

The proof is short once the right coordinates are chosen. In two dimensions, rotation has a single integer weight and there is a single radial variable. Thus every polynomial can be displayed schematically as

$$P = \sum_k \underbrace{T^k}_{\text{rotation}} \underbrace{B_k(U)}_{\text{radius}}.$$

Gaussian integration first takes the constant term in T, then replaces U^j by $j!$. If both positive and negative rotational weights occur, the lowest radial orders have a supporting line above weight zero. The polynomial carried by that lower face has a nonzero constant term in some power, by a theorem of Duistermaat and van der Kallen [2]. Raising that power to a suitable prime and reducing modulo the prime makes the lowest term impossible to cancel after the factorial substitution. This contradicts the assumed moment vanishing.

The argument is conceptual rather than computational. In particular, it contains the previously studied three-level and star-shaped supports without needing separate case analyses.

2 Circular coordinates

Let X, Y be independent standard real Gaussian variables. Set

$$Z = \frac{X + iY}{\sqrt{2}}, \qquad W = \frac{X - iY}{\sqrt{2}}.$$

Wick's formula gives

$$\mathbb{E}(Z^a W^b) = \begin{cases} a!, & a = b, \\ 0, & a \neq b. \end{cases} \tag{1}$$

Introduce a radial variable U and a Laurent variable T through the injective substitution

$$Z \longmapsto T, \qquad W \longmapsto UT^{-1}.$$

Thus $Z^a W^b$ becomes $T^{a-b}U^b$. Every polynomial $P(Z, W)$ therefore has a unique expression

$$P = \sum_{k \in S} T^k B_k(U), \tag{2}$$

where $S \subset \mathbb{Z}$ is finite and every $B_k \in \mathbb{C}[U]$ is nonzero. The integer k is the rotational weight.

Define the factorial functional

$$\mathcal{L} : \mathbb{C}[U] \longrightarrow \mathbb{C}, \qquad \mathcal{L}(U^j) = j!.$$

Equation (1) is exactly the identity

$$\mathbb{E}(F) = \mathcal{L}\big(\mathrm{CT}_T\, F\big) \qquad (F \in \mathbb{C}[Z, W]). \tag{3}$$

This is the feature special to two variables: there is one angular constant-term operation and one radial factorial.

3 The exposed lower face

For each $k \in S$, write

$$\nu_k = \mathrm{ord}_U\, B_k, \qquad b_k = [U^{\nu_k}]B_k \neq 0.$$

The points (k, ν_k) record the smallest radial degree at each rotational weight. We now extract the lower face visible from weight zero.

Lemma 3.1 (Lower face)**.** *Suppose* $0 \in \mathrm{conv}(S)$*. There are rational numbers* ρ, θ*, a subset*

$$S_0 = \{k \in S : \nu_k = \rho + \theta k\},$$

and an integer $r \geq 1$ with the following properties:

$$\nu_k \geq \rho + \theta k \quad (k \in S), \qquad 0 \in \operatorname{conv}(S_0),$$

and, for

$$P_0(T) = \sum_{k \in S_0} b_k T^k, \qquad F_m(U) = \mathrm{CT}_T(P^m),$$

there are $n \in \mathbb{Z}_{\geq 0}$ and $c \in \mathbb{C}^\times$ such that

$$\text{every term of } F_m \text{ has degree at least } \rho m, \tag{4}$$

$$n = \rho r, \qquad F_r(U) = cU^n + \text{terms of degree greater than } n. \tag{5}$$

Proof. Let

$$C = \operatorname{conv}\{(k, \nu_k) : k \in S\} \subset \mathbb{R}^2.$$

Since $0 \in \operatorname{conv}(S)$, the polygon C meets the vertical line $k = 0$. Let $(0, \rho)$ be the lowest point of this intersection. The lower boundary of C admits a supporting line through this point; write it as

$$\nu = \rho + \theta k.$$

Thus

$$\nu_k \geq \rho + \theta k.$$

The vertices of C are integral, so its lower edges have rational slopes. If $(0, \rho)$ is a vertex, choose a rational slope between the slopes of its adjacent lower edges; at an endpoint, choose a rational finite slope beyond the single adjacent slope. If C is a single point, take $\theta = 0$. Thus $\rho, \theta \in \mathbb{Q}$.

The supporting line cuts out a face of C, and this face contains $(0, \rho)$. Every point of the face is a convex combination of its vertices, which are among the points indexed by

$$S_0 = \{k \in S : \nu_k = \rho + \theta k\}.$$

Taking first coordinates in such a convex combination for $(0, \rho)$ gives $0 \in \operatorname{conv}(S_0)$.

We use the constant-term theorem of Duistermaat and van der Kallen [2]: if every positive power of a complex Laurent polynomial has zero constant term, then zero does not lie in the convex hull of its support. Applied contrapositively to P_0, it gives an $r \geq 1$ for which

$$c := \mathrm{CT}_T(P_0^r) \neq 0.$$

Consider a weight-zero product of m terms selected from (2). If the selected weights are $k_1, \ldots, k_m$, then its radial degree is at least

$$\sum_{a=1}^{m} \nu_{k_a} \geq \rho m + \theta \sum_{a=1}^{m} k_a = \rho m.$$

This proves (4). When $m = r$, equality can occur only by taking the lowest coefficient from weights on S_0, and the sum of all equality terms is c. Hence the lowest term of F_r is $cU^{\rho r}$. Its exponent is necessarily a nonnegative integer; call it n. $\square$

4 Filtered Frobenius isolation

For any characteristic-zero ring R, write

$$\mathcal{L}_R : R[U] \longrightarrow R, \qquad \mathcal{L}_R(U^j) = j!.$$

The arithmetic heart of the proof is independent of circular coordinates.

Lemma 4.1 (Filtered Frobenius isolation)**.** *Let R be a characteristic-zero domain, let κ be a field of characteristic $p > 0$, and let $R \to \kappa$, $a \mapsto \bar{a}$, be a ring homomorphism. Suppose $F_m \in R[U]$ satisfy, for some $\rho \in \mathbb{Q}$, $r \geq 1$, $n \in \mathbb{Z}_{\geq 0}$, and $c \in R \setminus \{0\}$,*

$$\operatorname{ord}_U F_m \geq \rho m, \qquad n = \rho r, \qquad F_r = cU^n + \text{terms of degree greater than } n.$$

If

$$\bar{c} \neq 0, \qquad \overline{F_{rp}} = \overline{F_r}^{\,p},$$

then the values $\mathcal{L}_R(F_m)$ cannot all vanish.

Proof. Suppose they do. The order bound at $m = rp$, together with $n = \rho r$, gives

$$\operatorname{ord}_U F_{rp} \geq \rho r p = np.$$

Thus $(np)!$ is a common factor of all factorial weights in $\mathcal{L}_R(F_{rp}) = 0$. It is nonzero in the characteristic-zero domain R, so it may be cancelled before reduction.

Write $F_r = cU^n + U^{n+1}G(U)$. After reduction, the Frobenius hypothesis gives

$$\overline{F_{rp}} = \overline{F_r}^{\,p} = \bar{c}^p U^{np} + U^{(n+1)p}\overline{G}(U)^p.$$

Writing the second term as $\sum_{j \geq n+1} a_j U^{jp}$, the cancelled moment identity reduces to

$$0 = \bar{c}^p + \sum_{j \geq n+1} a_j \frac{(jp)!}{(np)!}.$$

Every factorial quotient in the sum contains the factor $(n+1)p$, hence vanishes in characteristic p. Therefore $\bar{c}^p = 0$, contrary to $\bar{c} \neq 0$. $\square$

5 Proof of the theorem

Proof of Theorem 1.1. Write P as in (2) and suppose that $\mathbb{E}(P^m) = 0$ for every $m \geq 1$. By (3), this says

$$\mathcal{L}(F_m) = 0 \qquad (m \geq 1).$$

If $P = 0$, the conclusion is immediate, so assume $P \neq 0$. If $0 \in \operatorname{conv}(S)$, Lemma 3.1 produces a nonzero lowest coefficient c. Let $R \subset \mathbb{C}$ be the $\mathbb{Z}$-algebra generated by the coefficients of P and by c^{-1}. Choose a maximal ideal $\mathfrak{m} \subset R$. Its residue field κ is finite by the weak Nullstellensatz for finite-type $\mathbb{Z}$-algebras; write $p = \operatorname{char}\kappa$. The image of c is nonzero because c is a unit in R. Moreover, in characteristic p,

$$\overline{F_{rp}} = \operatorname{CT}_T\big((\bar{P}^r)^p\big) = \big(\operatorname{CT}_T(\bar{P}^r)\big)^p = \overline{F_r}^{\,p},$$

since Frobenius sends T^i to T^{pi}. Lemma 4.1, with the bounds from Lemma 3.1, now gives a contradiction. Therefore

$$0 \notin \operatorname{conv}(S).$$

Since $S \subset \mathbb{Z}$, either every weight of P is strictly positive or every weight is strictly negative.

Now fix a polynomial Q. If $Q = 0$, there is nothing to prove, so let S_Q be its nonempty finite set of rotational weights. Suppose first that all weights of P are positive, and put

$$a = \min S > 0, \qquad b = \min S_Q, \qquad N = \max\left\{0, \left\lfloor \frac{-b}{a} \right\rfloor + 1\right\}.$$

Every weight of QP^m is at least $b + ma > 0$ for $m \geq N$.

If instead all weights of P are negative, put

$$a = -\max S > 0, \qquad b = \max S_Q, \qquad N = \max\left\{0, \left\lfloor \frac{b}{a} \right\rfloor + 1\right\}.$$

Then every weight of QP^m is at most $b - ma < 0$ for $m \geq N$. Thus in either case $\mathrm{CT}_T(QP^m) = 0$ for every $m \geq N$. Equation (3) then gives $\mathbb{E}(QP^m) = 0$, as required. $\square$

Corollary 5.1. *The Gaussian Moments Conjecture holds in dimension d exactly when $d \leq 2$.*

Proof. The case $d = 1$ is known [1, Proposition 4.2]. Theorem 1.1 proves the case $d = 2$. Long's examples disprove the conjecture for every $d \geq 3$ [3]. $\square$

6 What the argument says

The proof gives a stronger structural statement than the mixed-moment conclusion alone:

$$\mathbb{E}(P^m) = 0 \text{ for all } m \geq 1 \quad \Longrightarrow \quad \text{all rotational weights of } P \text{ have one strict sign.}$$

Thus vanishing in two variables is not caused by a complicated balance among positive, zero, and negative angular modes. Such a balance always has a lowest radial face, and a prime moment detects it.

This also explains why support-by-support calculations quickly appeared to show rigidity. A three-level support, a star support, or a tree of primitive weight relations is only a finite presentation of the same lower face. Leaf removal is unnecessary: the supporting line exposes all relevant leaves at once.

The proof is genuinely two-dimensional. With one circular pair there is one weight coordinate and one radial order, and the latter is measured by the single factorial $j!$. In higher dimension there are several radial directions, so the one-dimensional isolation step no longer applies. Corollary 5.1 shows that this is not merely a technical gap: counterexamples already occur in dimension three.

7 Formal verification

A companion Lean 4 development has been prepared as supplementary material for the two-variable theorem. Its exported bivariate statement starts with an ordinary polynomial in two variables, verifies the circular substitution and Wick monomial formula, and proves the full mixed-moment conclusion from vanishing of all positive pure moments. Internally, it checks rational supporting-face extraction, the weighted associated-graded identification of the lowest radial coefficient, descent to the finite coefficient $\mathbb{Z}$-algebra, normalized factorial cancellation, Frobenius and constant-term commutation, prime-dilated factorial isolation, and the eventual vanishing argument for both signs of the rotational support. The development contains no `sorry` declarations.

The formalization exposes two external inputs as named axioms: the Duistermaat–van der Kallen constant-term theorem and the existence of good finite-characteristic specializations of finite-type $\mathbb{Z}$-domains. The rational supporting face is proved directly from the finite two-dimensional polygon. An axiom audit of the exported theorem therefore lists precisely these two mathematical inputs (besides Lean's standard logical quotient and choice principles). Thus the artifact gives a machine-checked verification of the complete proof of Theorem 1.1, with its remaining trust boundary stated explicitly.

Acknowledgments and declaration of generative AI use

The author used OpenAI's ChatGPT and Codex extensively during the research process, including for conjecture formation, symbolic experimentation, proof exploration, consistency checks, and preparation of this manuscript. The lower-face and prime-isolation argument emerged from that collaborative process. The author reviewed the mathematical claims and assumes full responsibility for the contents. The proof of Theorem 1.1 is noncomputational.

References

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